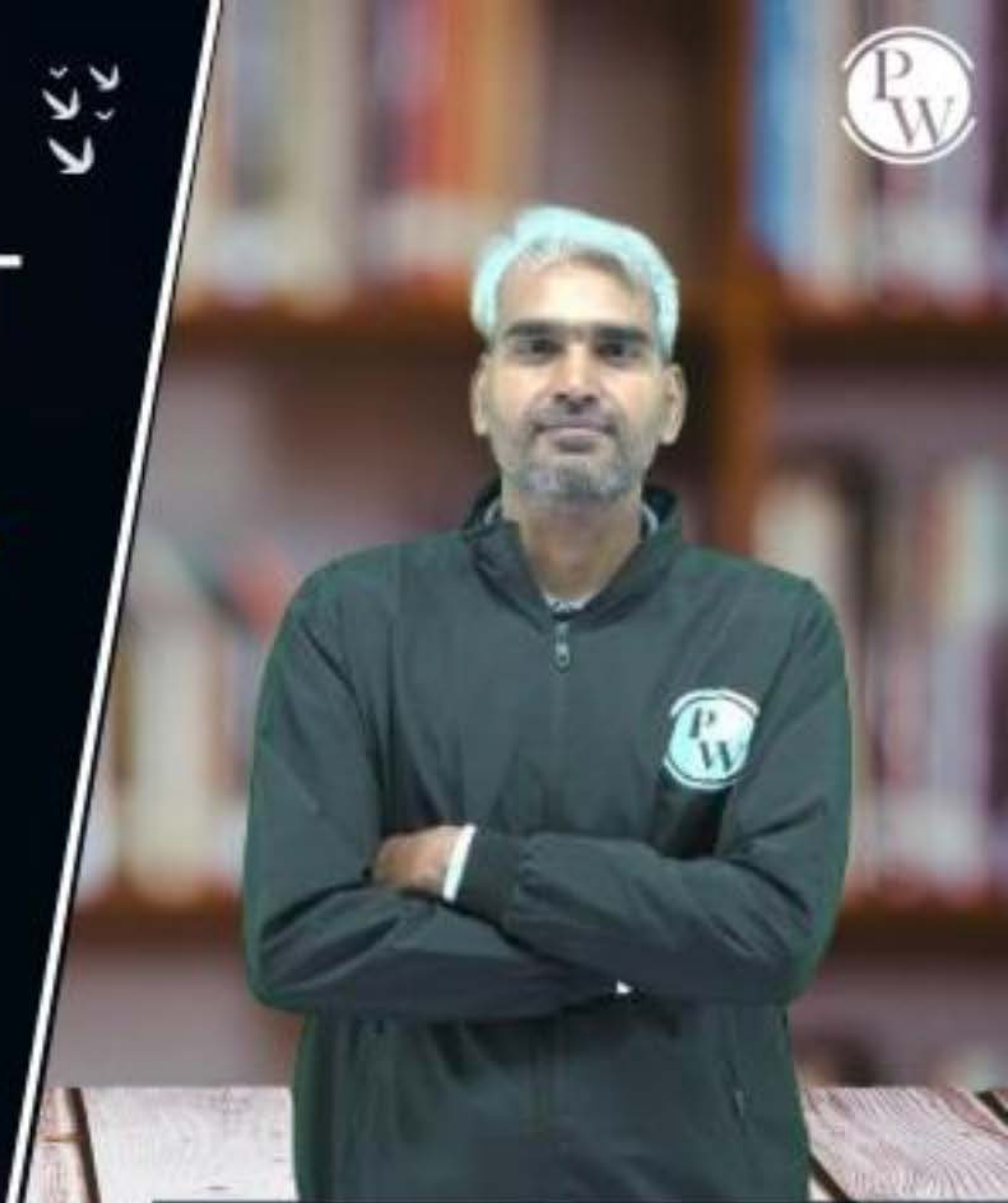
Computer Science & IT

Database Management System

Query Languages

Lecture No. 09



By- Vishal Sir

Recap of Previous Lecture

- **Topic** IN, ANY, ALL and EXISTS operators
- **Topic** Execution w.r.t. correlated subquery

Slide

Topics to be Covered

- **Topic** Execution w.r.t. correlated subquery
- **Topic** AS clause
- **Topic** WITH clause
- **Topic** Comparison with NULL

Slide

What output will be produce by the following query On the relations

Catalog

Sid	Pid
S_1	P_1
S_1	P_2
S_3	P_2
S_3	P_3

Parts

Pid	Color
P_1	Red
P_2	Green
P_3	Red

#e.g.

```
SELECT C.sid
FROM Catalog C
WHERE EXISTS ( SELECT *
               FROM Parts P
               WHERE (P.Pid=C.Pid AND P.color='RED'))
```

```
Select C.Sid
  from Catalog C
    Where EXISTS ( Select *
                    from Parts P
                    Where (C.Pid = P.Pid AND P.Color = 'Red'))
```

C

① S_1 P_1

o/p ⇒ P_1 Red

Parts

Pid	Color
P_1	Red
P_2	Green
P_3	Red

Not Empty

∴ EXISTS return True

↓

∴ Where Condn is true

∴ 1st tuple of Catalog is selected

Catalog

	Sid	Pid
1	S_1	P_1
2	S_1	P_2
3	S_3	P_2
4	S_3	P_3

→ S_1 P_1

② S_1 P_2

```
Select C.Sid
    from Catalog C
        Where EXISTS ( Select *  {Empty}
                          from Parts P
                          Where (C.Pid = P.Pid AND P.Color = 'Red'))
```

∴ False
↓
Hence 2^{nd} tuple of Catalog is not selected.

Parts

Pid	Color
P_1	Red
P_2	Green
P_3	Red

Catalog

	Sid	Pid
1	S_1	P_1
2	S_1	P_2
3	S_3	P_2
4	S_3	P_3

S_1 P_1 ✓

$\boxed{S_1 \; P_2}$ ✗

③ S_3 P_2

Select C.Sid
from Catalog C
Where EXISTS (Select * {Empty}
from Parts P
Where (C.Pid = P.Pid AND P.Color = 'Red'))

∴ False

∴ 3rd tuple is not selected.

Parts

Pid	Color	
P_1	Red	✗
P_2	Green	✗
P_3	Red	✗

Catalog

	Sid	Pid		
①	S_1	P_1	S_1 P_1	✓
②	S_1	P_2	$\boxed{S_1\ P_2}$	✗
③	S_3	P_2	$\boxed{S_3\ P_2}$	✗
④	S_3	P_3		

④ S3 P3

```
Select C.Sid
  from Catalog C
    Where EXISTS ( Select * {(P3, Red)}
                     From Parts P
                     Where (C.Pid = P.Pid AND P.Color = 'Red'))
```

Not Empty
∴ True
Hence 4th tuple is selected

Parts

Pid	Color
P_1	Red ✗
P_2	Green ✗
P_3	Red ✓

Catalog

	Sid	Pid
①	S_1	P_1
②	S_1	P_2
③	S_3	P_2
④	S_3	P_3

S_1 P_1 ✓

$\boxed{S_1 \; P_2}$ ✗

$\boxed{S_3 \; P_2}$ ✗

S_3 P_3 ✓

From the selected tuples result Sids in o/p ⇒ ∴ o/p =

Sid
S_1
S_3

✓

H.W.

#e.g.

```
SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS ( SELECT P.Pid
                   FROM Parts P
                   WHERE NOT EXISTS (SELECT C2.Sid
                                     FROM Catalog C2
                                     WHERE(C2.Pid=P.Pid AND  C2.Sid=C1.Sid)))
```

Catalog

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2

Parts

Pid	Color
P_1	Red
P_2	green

What output is produced by above SQL query:

A. $\boxed{S_1, S_2}$ Sids of suppliers who supplied some parts {at least one part}

B. $\boxed{S_2}$ Sids of suppliers who supplied only proper subset of parts from all parts

C. $\boxed{S_1}$ Sids of suppliers who supplied all parts

D. Sids of suppliers who did not supply any part {Not Possible}

S_1 P_1

```
SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS ( SELECT P.Pid
                   FROM Parts P
                   WHERE NOT EXISTS (SELECT C2.Sid
                                     FROM Catalog C2
                                     WHERE(C2.Pid=P.Pid AND C2.Sid=C1.Sid)))
```

C_1

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2

P

Pid
P_1
P_2

C_2

Sid	Pid	
S_1	P_1	✓
S_1	P_2	×
S_2	P_2	×

P_1 is not selected

S_1 P_1 is selected

$\therefore$ Not exists returns false

↓

hence

SELECT C1.sid hence "Sid" from $\boxed{S_1 \; P_1}$ is selected

$S_1 \; P_1$

FROM Catalog C1

WHERE NOT EXISTS (SELECT P.Pid

FROM Parts P

WHERE NOT EXISTS (SELECT C2.Sid

FROM Catalog C2

WHERE(C2.Pid=P.Pid AND C2.Sid=C1.Sid)))

C_1

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2

P

Pid
P_1
P_2

C_2

Sid	Pid	
S_1	P_1	×
S_1	P_2	✓
S_2	P_2	×

$\cancel{P_1}$ $\cancel{P_2}$ is not selected

Inner query result is Empty

$\therefore$ NOT EXISTS will return true

$S_1 \; P_2$ is selected

$\therefore$ NOT EXISTS returns false

and hence

S_1 P_2

```
SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS ( SELECT P.Pid
                   FROM Parts P
                   WHERE NOT EXISTS (SELECT C2.Sid
                                     FROM Catalog C2
                                     WHERE(C2.Pid=P.Pid AND  C2.Sid=C1.Sid)))
```

~~P_1~~ not selected

return False

C_1

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2

P

Pid
P_1
P_2

C_2

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2

```
SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS ( SELECT P.Pid
    FROM Parts P
    WHERE NOT EXISTS (SELECT C2.Sid
        FROM Catalog C2
        WHERE(C2.Pid=P.Pid AND  C2.Sid=C1.Sid)))
```

C_1

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2

S_1	P_2

Hence Sid from 2^{nd} tuple is also selected

Empty ∴ Return true

P

Pid
P_1
P_2

~~P_1~~ ∴ P_2 is also not selected

return False

C_2

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2

$\boxed{S_2 \quad P_2}$

```
SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS ( SELECT P.Pid
                   FROM Parts P
                   WHERE NOT EXISTS (SELECT C2.Sid
                                     FROM Catalog C2
                                     WHERE(C2.Pid=P.Pid AND  C2.Sid=C1.Sid)))
```

C_1

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2

P

Pid
P_1
P_2

Pid
P_1 is selected

C_2

Sid	Pid	
S_1	P_1	✗
S_1	P_2	✗
S_2	P_2	✗

Empty
↓
∴ Returns True
↓
∴

```
SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS ( SELECT P.Pid
    FROM Parts P
    WHERE NOT EXISTS (SELECT C2.Sid
        FROM Catalog C2
        WHERE(C2.Pid=P.Pid AND  C2.Sid=C1.Sid)))
```

C1

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2

$\boxed{S_2 \quad P_2}$

where return false w.r.t. S_2

P

Pid
P_1
P_2

C2

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2

Pid P_1 is selected

Empty ∴ Returns True ∴

∴ NOT EXISTS will return false and hence Sid of 3rd tuple will not be selected

#e.g.

```
SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS ( SELECT P.Pid
                   FROM Parts P
                   WHERE NOT EXISTS (SELECT C2.Sid
                                     FROM Catalog C2
                                     WHERE(C2.Pid=P.Pid AND  C2.Sid=C1.Sid)))
```

Catalog

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2

Selected → S_1

Not selected → S_2

Parts

Pid	Color
P_1	Red
P_2	green

$\boxed{o/p = S_1}$

What output is produced by above SQL query:

A. Sids of suppliers who supplied some parts {at least one part}

B. Sids of suppliers who supplied only proper subset of parts from all parts

C. Sids of suppliers who supplied all parts ✓

D. Sids of suppliers who did not supply any part {Not Possible} if we use distinct, then No. of tuples in o/p = 2

```
SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS ( SELECT P.Pid
                   FROM Parts P
                   WHERE NOT EXISTS (SELECT C2.Sid
                                     FROM Catalog C2
                                     WHERE(C2.Pid=P.Pid AND  C2.Sid=C1.Sid)))
```

If the given query is executed over the following instances of Catalog and Parts, the how many tuples will be present in o/p

Catalog

Sid	Pid
S1	P1
S1	P2
S2	P2
S3	P1
S3	P2

4 tuples in o/p

Parts

Pid
P1
P2

Topic : AS clause

{ Using 'AS' clause we can rename almost every thing }

AS clause is used to rename a column (attributes) or table with an alias.

An alias only exists for the duration of the query execution.

#e.g. Consider a database that has the relation schema EMP (EmpId, EmpName, and DeptName).
An instance of the schema EMP and a SQL query on it are given below:

EMP

EmpId	Emp Name	DeptName
1	XYA	AA
2	XYB	AA
3	XYC	AA
4	XYD	AA
5	XYE	AB
6	XYF	AB
7	XYG	AB
8	XYH	AC
9	XYI	AC
10	XYJ	AC
11	XYK	AD
12	XYL	AD
13	XYM	AE

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
    (SELECT DeptName, COUNT(EmpId) AS EC(DeptName, Num)
                    FROM EMP
                    GROUP BY DeptName)
```

The output of executing the SQL query is__________

#e.g. Consider a database that has the relation schema EMP (EmpId, EmpName, and DeptName).
An instance of the schema EMP and a SQL query on it are given below:

EC(DeptName, NUm)

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
    (SELECT DeptName, COUNT(EmpId) AS EC(DeptName, Num)
     FROM EMP
     GROUP BY DeptName)
```

The output of executing the SQL query is ________

EC

O/p without "AS" will be

DeptName DeptName	Num Count (EmpID)
AA	4
AB	3
AC	3
AD	2
AE	1

it is an independent sub-query ⇒ ∴ it is executed first, because of its execute table EC(DeptName, NUm) is visualized.

this O/p is renamed AS EC(DeptName, Num)

i.e. the O/p will be treated as relation EC

1st attribute is renamed as DeptName, and 2nd one as NUm

#e.g. Consider a database that has the relation schema EMP (EmpId, EmpName, and DeptName).
An instance of the schema EMP and a SQL query on it are given below:

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
    (SELECT DeptName, COUNT(EmpId) AS EC(DeptName, Num)
                                    FROM EMP
                                    GROUP BY DeptName)
```

The output of executing the SQL query is ________

after execution of inner query, our outer query will execute
outer query will execute on EC table itself

o/p of where clause of outer query

DeptName	Num
AA	4
AB	3
AC	3
AD	2
AE	1

on this o/p of where Aggregate function will be applied

$$\text{Avg (Num)} = \frac{\text{Sum(Num)}}{\text{Count (Num)}} = \frac{13}{5} = 2.6$$

Ans

o/p

EC

DeptName	Num
AA	4
AB	3
AC	3
AD	2
AE	1

{"With" is used along with "As"}

Topic : WITH clause

The WITH Clause is mainly used to provide a subquery block

[redacted] a name that can be referenced within the main SQL query. or other subquery that follows.

o/p of subquery

Syntax of WITH:

Here we will define a new schema

Here we have a sub-query

WITH New-relⁿ_name(, , ...,) AS Sub-query block

list of new names for the attributes

The output of this subquery will be assigned a new-name by 'WITH' clause

We may have multiple such sub-query with "WITH" clause.

In the end we will have our main query

Select - - -

- - - -

- - -

our main query

#e.g. Consider the following database table named water_scheme

water_scheme		
scheme_no	district_name	capacity
1	Ajmer	20
1	Bikaner	10
2	Bikaner	10
3	Bikaner	20
1	Churu	10
2	Churu	20
1	Dungargarh	10

Order of execution

The number of tuples returned by the following SQL query is__________

```
① with total(name, capacity) as
        select district_name, sum(capacity)
        from water_schemes
        group by district_name
② with total_avg(capacity) as
        select avg(capacity)
        from total
③ select name from total, total_avg
        where total.capacity >= total_avg.capacity
```

Main query

#e.g. Consider the following database table named water_scheme

total

name	Capacity

O/P

Dist.name	Sum(capacity)
Ajmer	20
Bikaner	40
Churu	30
Dungargarh	10

New schema w.r.t. o/p of subquery

The number of tuples returned by the following SQL query is ______

subquery block

1

```
with total(name, capacity) as
        select district_name, sum(capacity)
        from water_schemes
        group by district_name
```

2

```
with total_avg(capacity) as
        select avg(capacity)
        from total
```

3

```
select name from total, total_avg
        where total.capacity >= total_avg.capacity
```

total_avg

Capacity
Avg(capacity)
25

(total x total_avg) =

total		total_avg
name	Capacity	Capacity
A	20	25
B	40	25
C	30	25
D	10	25

'2' tuples in O/P

H.W.

~~#Q.~~ Consider the relational database with the following four schemes and their respective instances.

Student(sNo, sName, dNo) Dept(dNo, dName)

Course(cNo, cName, dNo) Register(sNo, cNo)

Student

sNo	sName	dNo
S01	James	D01
S02	Rocky	D01
S03	Jackson	D02
S04	Jane	D01
S05	Milli	D02

Dept

dNo	dName
D01	CSE
D02	EEE

Course

cNo	cName	dNo
C11	DS	D01
C12	OS	D01
C21	DE	D02
C22	PT	D02
C23	CV	D03

Register

sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

Question Continues in Next Slide

#Q. SQL Query:

```
SELECT * FROM Student AS S
WHERE NOT EXIST
        (SELECT cNo FROM Course WHERE dNo = "D01".
                EXCEPT ≡ (Minus)
        SELECT cNo FROM Register WHERE sNo = S.sNo)
```

The number of rows returned by the above SQL query is ______.

2 mins Summary

Topic Order of Execution of Correlated Sub-query.

Topic "AS" Clause

"WITH" Clause

Slide

THANK - YOU

Slide